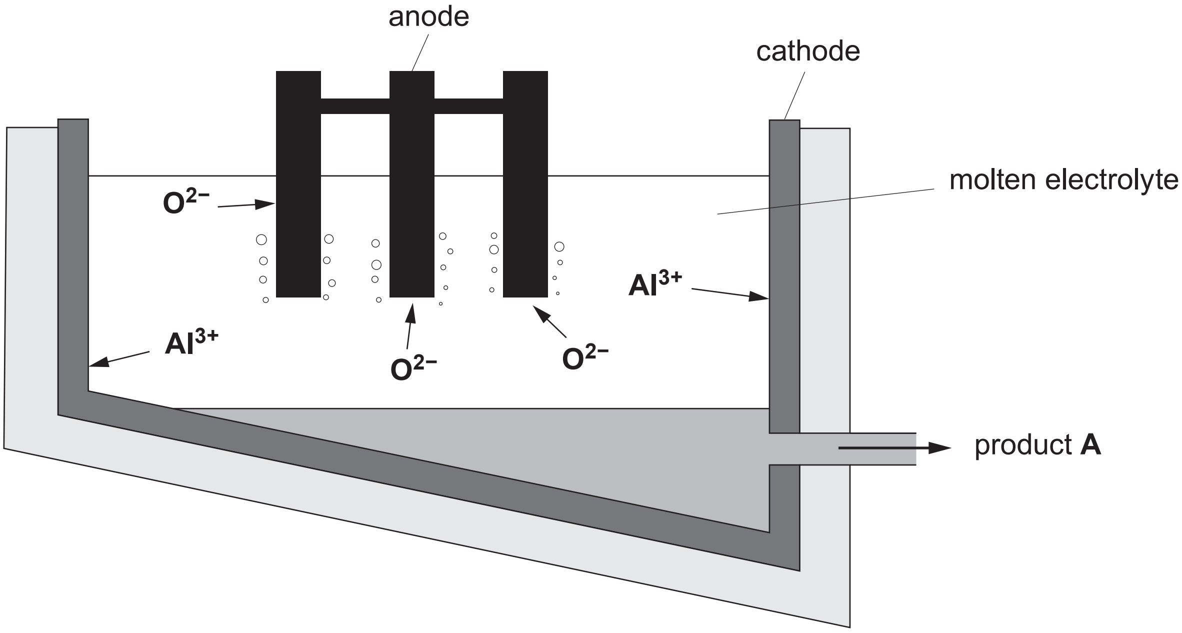
Examiner only

2. The diagram shows an electrolysis cell used in the extraction of aluminium from alumina.



bauxite	**electrical**	**molten alumina**	**positive**	
negative	**cryolite**	**molten aluminium**	**oxygen**	**light**

(a) Choose words from the box to complete the following sentences. [5]

The molten electrolyte is a mixture of alumina and ....

Product **A** is ....

Oxygen is formed at the anode which is the .... electrode.

Alumina is obtained from an ore called ....

The process of extracting aluminium is expensive because it uses a lot of

.... energy.

(b) Balance the symbol equation that shows the overall reaction. [1]

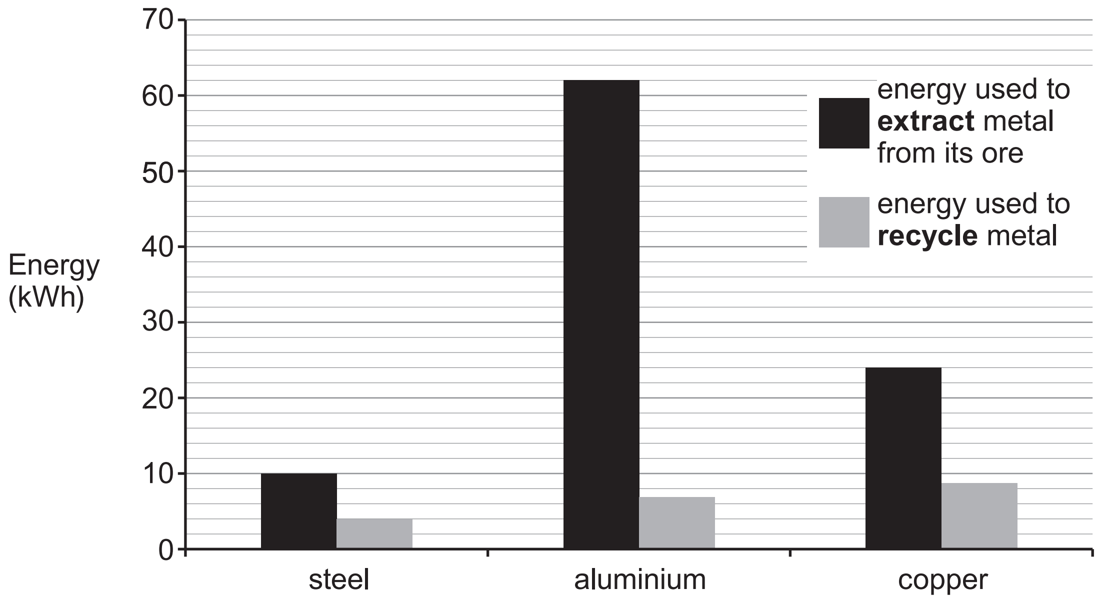
$$2Al_2O_3 \longrightarrow \boxed{}\ Al + 3O_2$$

Examiner only

(c) 250 tonnes of an aluminium ore can produce 195 tonnes of alumina.
Calculate the percentage of alumina in the ore. [2]

Percentage = ... %

(d) The chart compares the amount of energy used to extract and to recycle three metals.



Use the information in the chart to answer the following question.

Put a tick (✓) in the box next to the statement that describes why it is more cost effective to recycle aluminium than steel and copper. [1]

The energy used to extract metals is greater than that used in recycling them ☐

The difference between the energy used to extract and the energy used to recycle is the greatest ☐

The energy used in recycling is less than for copper but greater than for steel ☐

9

3410U201 07